

EngTrace — Annotation Guide

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1. Your Annotation File

You will receive one JSONL file per model. Each line is a JSON object representing one trace — a confirmed case where the model produced a wrong answer. The fields are:

Field		Description
annotation_id	Read-only	Unique ID for this trace.
model	Read-only	Model name.
question_id	Read-only	Original benchmark question ID.
difficulty	Read-only	Easy / Intermediate / Advanced.
branch	Read-only	Engineering branch (chemical / electrical / mechanical).
domain	Read-only	Engineering domain, e.g. thermodynamics.
problem_statement	Read-only	The full problem text. Read this first.
gold_answer	Read-only	The correct reference solution. Use as ground truth.
model_reasoning	Read-only	The model's full chain-of-thought. This is what you annotate.
final_answer_acc	Read-only	0.0 for every row — all traces are confirmed wrong answers.
recovered_f1	Read-only	Step-level alignment score. Informational only.
recall	Read-only	Step-level recall. Informational only.
tribunal_triggered	Read-only	false for every row.
decision_step_triggered	YOU FILL	Q1, Q2, Q3, Q4, Q5, or Q6 — the first question that answered YES.
error_category	YOU FILL	Integer 1–6.
error_category_name	YOU FILL	Full name, e.g. 'Formula / Principle Error'.
triggering_text_span	YOU	Short verbatim excerpt from model_reasoning that shows the error.

Field		Description
	FILL	
annotator_id	YOU FILL	Your name or ID.
annotator_notes	YOU FILL	Optional. Flag anything ambiguous or unusual.
adjudicated	Leave blank	
final_label	Leave blank	

What to fill in: For each trace, read the **problem_statement**, compare the **model_reasoning** against the **gold_answer**, then fill in the four highlighted fields: **decision_step_triggered**, **error_category**, **error_category_name**, and **triggering_text_span**.

triggering_text_span: Copy a short excerpt — one sentence or one equation — from **model_reasoning** that directly shows the error. This is the evidence for your label.

2. Error Categories

Assign exactly one category per trace. The categories are ordered by severity — errors higher in the list are more fundamental and take priority.

#	Category	Definition	Severity
1	Hallucination	Model uses a value, constant, or equation that does not exist in any real engineering reference — it was fabricated.	Critical
2	Setup / Assumption Error	Problem is framed incorrectly before any equation is applied — wrong boundary condition, ignored stated constraint, or misidentified system configuration.	High
3	Formula / Principle Error	Setup is correct but the wrong real governing equation, law, or principle is selected.	High
4	Unit / Dimensional Error	Correct formula and setup, but dimensional consistency fails — missing conversion, mixed unit systems, or value substituted in wrong unit.	Medium
5	Sign / Direction Error	Correct formula, units, and setup — but wrong sign or physical direction assigned to a quantity, violating the reference frame or process direction.	Medium
6	Calculation Error	Everything is correct — setup, equation, units, sign — but the arithmetic or algebraic execution produces the wrong number.	Low

3. How to Assign a Category

Go through the questions below in order. Answer YES or NO to each. Stop at the first YES and assign that category. Do not continue to lower questions once you have a YES.

Step	Question — answer in order, stop at first YES	If YES
Q1	Does the model use a value, constant, or equation that does not exist in any real engineering reference?	→ Hallucination (1)
Q2	Is the problem framed incorrectly before any equation is applied? (wrong boundary, ignored constraint, misidentified configuration)	→ Setup / Assumption Error (2)
Q3	Is the selected governing equation or principle wrong (given the setup is correct)?	→ Formula / Principle Error (3)
Q4	Does dimensional analysis fail — missing conversion, mixed units, or value in wrong unit?	→ Unit / Dimensional Error (4)
Q5	Is the sign or physical direction of any quantity wrong (given units and formula are correct)?	→ Sign / Direction Error (5)
Q6	Does arithmetic or algebraic execution produce the wrong number?	→ Calculation Error (6)

Why this order: A higher-level error invalidates everything below it. A model that fabricates an equation has not meaningfully 'calculated' anything — labelling it a Calculation Error would be misleading. Always assign the first matching category.